

Memory Management Strategies for Maintaining Long-Term Dialogue Coherence and Personalization in Generation Chatbots

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Abstract—Generation-based chatbots, which read and write natural language and create new sentences, have advanced significantly, through big language models (BLMs). Nevertheless, the difficulty of reaching long-term coherence and a successful personalization is another challenge because of the shortcomings in managing memories. This work is a detailed study of memory systems, short-term memory (STM), long-term memory (LTM), episodic memory (EM), semantic memory (SM) and their contributions to continuity of context in dialogue systems. It points out fundamental problems of contextual drift, catastrophic forgetting (CF) and scale problems. Also, the paper examines personalization methods such as user profiling (UP), dynamic embeddings (DE), context-aware generation (CAG), and continual learning (CL) and privacy-preserving methods like federated learning (FL). This work will contribute to the further improvement of intelligent, memory- and user-adaptive conversation agents by studying the existing approaches and hypothesizing the integrated ones. The research also involves a comparative analysis of the latest contributions to research and future recommendations to adaptability and human-like interactions in chatbot systems. Such knowledge is critical in the development of chatbots that can remember the user context over time as well as adapt to user preferences. This study serves as a foundation for developing more emotionally intelligent and context-aware dialogue systems.

Keywords—Memory management, long-term memory retention (LTM), memory-augmented neural networks, retrieval-based memory, personalized chatbots, federated learning.

I. INTRODUCTION

A chatbot—also referred to as a chatter bot—is a piece of software that uses natural language processing (NLP) and artificial intelligence (AI) to have text or voice conversations with people. Chatbots are found in many industries and used for various reasons including automated customer service, virtual assistant, and information seeking[1]. As Frank noted in the past, the opportunity for repetition of human interaction increased with AI conditions. Moreover, with increased access to processing power and general open-source frameworks, the process of building a chatbot has become more common, and uptake has become quicker in corporate and academia.

The conversational AI market is growing exponentially. Market Research Future projects that the conversational AI market reach USD 32.5 billion by 2030, growing at a compound annual growth rate (CAGR) of 22.6% between 2022 and 2030[2]. The market is also expected to grow from USD 10.7 billion in 2023 to USD 29.8 billion by 2028, according to Markets and Markets. The growth in conversational AI is a byproduct of advancement in deep learning, transformer-based architectures, and necessity for

scalable, intelligent digital interactions[3]. As chatbots become more ubiquitous there greater demand for chatbots to have natural, coherent, and personalized dialog over longer conversations.

While recent improvements in conversational capabilities are commendable, generation-based chatbots struggle with appropriate long-term dialogue coherence and personalization. Issues of incoherence arise due to sporadic memory management, including issues with loss of contextual understanding, limited emotional intelligence, and inconsistent personality[4]. The majority of generation-based systems are required to maintain a list of previous conversational states. Therefore, the systems tend to default to repetitive, generic, and/or contradictory references to previous dialogue[5]. The negative user experience is compounded by various forms of gender bias or the inability to effectively interpret implications related to the overall user intent. Each of these attributes suggest a clear gap in overall memory strategies being able to establish the necessary contextualized dependability required to prolong a conversation and to incorporate individual profile/user needs.

This study looks into and investigates memory management strategies that achieve long-term coherence and personalization in generation-based chatbots[6]. discuss architectures and mechanisms such as memory-augmented neural networks, retrieval-based approach, knowledge graphs, and continual learning, that can remember and subsequently leverage prior interactions[7]. A systematic investigation of these approaches informs the pursuit of intelligent, coherent, and user-adaptive conversational agents that partake in on-going realistic human-like interactions.

A. Structure of the paper

This paper explores memory management in generation chatbots. Section II covers foundational memory architectures, while Section III discusses challenges in maintaining long-term coherence. Section IV presents personalization techniques such as user profiling and continual learning. Section V examines relevant literature, while Section VI offers important takeaways and suggestions for further study.

II. FUNDAMENTALS OF MEMORY MANAGEMENT IN DIALOGUE SYSTEMS

A. Overview of Conversational AI and Generation Chatbots

Conversational Artificial Intelligence (AI) is defined as a set of technologies that allow machines to converse with people in a human-like way through natural language understanding, processing, and generation. The components

can vary, but at a minimum there are some combination of: Natural language processing (NLP), response creation, dialogue management, and speech recognition[8]. Within the conversational agent taxonomy, generation-based chatbots are an advanced and sophisticated type capable of generating responses to the input in real-time rather than using a defined script or a retrieval-based non-scripted method to generate responses. Large, previously learnt language models like GPT and BERT are used by generation-based chatbots, or T5 to generate a response that is coherent and contextually relevant to the input, that closely follows human conversation patterns. Notably, generation-based chatbots can dynamically generate responses given information about the conversation context, unlike rule-based or retrieval-based systems[9]. The generation capabilities of these types of bots enable greater adaptability and flexibility than other forms of chatbots which allows for open-domain conversations and personalized interactions. Another example of where generation-based chatbots could be used is providing intelligent assistance to users, like recommending media. In addition to the adaptability and flexibility offered by the generative functions of model-based bots, challenges remain around maintaining a coherent dialog, handling hallucinations, and determining alignment with user intents, especially over the long-term duration of conversation. Consequently, it is important to think about using memory management strategies that can improve coherence and personalization of chatbot models.

B. Role of Memory in Conversational Contexts

Memory is a fundamental building block for coherent and contextually relevant conversations in AI dialogue systems. Memory allows interlocutors in human interaction to remember past interactions, build on them, stay within topics, adapt to changes in context, and personalize interactions based on prior knowledge[10]. Memory allows conversational agents to store, retrieve, and utilize information from previous dialogue turns, and if so, to maintain continuity, avoid repetitiveness, and maintain consistent behavior. A mechanism of memory is responsible for tracking user inputs in a generation-based chatbot, managing dialogue history, and preserving contextual dependencies across conversations and sessions. Without memory management[11], chatbots tend to give generic and sometimes contradictory responses for forgetting the previous conversations and user dialog preferences. This not only reduces the overall quality of the conversation; it also limits the overall opportunities for personalization. To achieve human-like dialogue competence, need to implement a structured memory architecture(s) and a forward or backward way of integrating reference to past conversation context. Memory would allow the system to encode, retrieve and updated conversational information related to the conversation, in real time. Memory forms the basis for long-term coherence, accomplish tasks, and allow users to trust intelligent dialogue systems.

C. Types of Memory Architectures

Memory architectures represent an important function to store dialogue context, user preferences, and prior interactions for conversational AI systems, particularly generation-based chatbots[12][13]. Memory architectures come from cognitive theory of human memory and can be implemented in AI systems to allow improved contextual awareness, coherency, and personalization. Memory may be separated into episodic and semantic memory, as well as short-term and long-term memory, all of which provide different functions to model the

dialogue, also comparative discussions of Memory Architectures in Generation Chatbots are mentioned in table 1.

1) Short-Term Memory

Short-term memory (STM) in dialogue systems is defined as the conversational context established during conversational turns as parts of a dialogue system's interactions. STM typically maintains information about the current dialogue turn, a small history of recent utterances, and short-lived conversational state. It is fleeting in nature and is refreshed or erased at the end of a session, or after a small number of conversational turns. In neural models, short-term memory is generally implemented through recurrent neural networks (RNNs), attention schemes and intermediate systems of dialogue state tracking systems which capture contextual dependencies temporarily[14]. Also, while STM has a critical role optimizing relevant and coherent responses to utterances in short dialogues, it is not sufficient for keeping cohesive conversations during extended interactions or ongoing sessions. Thus, improving short-term memory with formal representations and dynamic updating is an active area of research in conversational AI.

2) Long-Term Memory

Chatbots are able to remember and recall information in long-term memory (LTM) over a range of discussions and durations. LTM allows the chatbots to continually refresh the user preferences, past dialogues, user intentions, and outside knowledge to interact with the user on a more personalized and steadier base. LTM is usually realized as memory-augmented neural networks or as external knowledge bases or embedding-based retrieval systems[15]. In contrast to STM, in which the primary action is within short-term time ranges, LTM can enable temporal continuity as well as the relevance of information within longer-term time ranges. By introducing the LTM into the chatbot, it will enable that chatbot to behave more human-like in machine format able to engage in unique learning through its prior experiences with a given user, not repeat the same interactions, and finally formulate user-specific conversational contexts in each interaction. Last, LTM is complex when it comes to storage of memory, efficiency in calculations, privacy, and data update standards.

3) Episodic vs. Semantic Memory

Episodic and semantic memories are two variants of the same but complementary memory by the science of cognition and the growing holistic participation in the conversational agents. The physical record of particular past events or interaction like the previous prompt or dialogue of user, is known as episodic memory, and explains the rooted conversation in the past. The episodic memory incorporates time and context of former conversations that enable individual and accrued reactions[16]. Semantic memory, in its turn, is the repository of generalized knowledge about the world and knowledge of concepts and language patterns to which the chatbot resorts in order to comprehend and respond in many different areas. Whereas episodic memory is changeable and person-specific, semantic memory is more fixed and general. Chatbots can be used to hold factually-based yet contextually-informed conversations that can be achieved by integrating episodic and semantic memory. The proportion between episodic and semantic types of memory is an important requirement so that on the long term, a

TABLE I. COMPARATIVE OVERVIEW OF MEMORY ARCHITECTURES IN GENERATION CHATBOTS

Aspect	Short-Term Memory (STM)	Long-Term Memory (LTM)	Episodic Memory	Semantic Memory
Definition	Maintains recent dialogue turns and context	Stores and retrieves information across sessions	Stores records of specific past events or conversations	Stores generalized world knowledge and language patterns
Lifespan	Temporary (within a session or few turns)	Persistent across sessions and long durations	Dynamic, event-specific and user-centric	Static or slowly updated over time
Purpose	Ensures contextual coherence in short conversations	Supports continuity, personalization, and goal-tracking	Anchors conversations with time/context awareness	Provides background knowledge and linguistic understanding
Implementation Techniques	RNNs, attention mechanism, dialogue state tracking	Memory-augmented neural networks, external KBs, embedding retrieval	Time-stamped or session-specific memory modules	Pretrained language models, ontologies, and conceptual graphs
Scope	Focused on immediate past exchanges	Encompasses wide historical and cross-session data	Specific past user interactions (e.g., last query or feedback)	Domain-independent knowledge (e.g., facts, grammar, word meaning)
Contribution to Personalization	Limited personalization, mostly session-based	Enables deep personalization and user modeling	Allows accumulation of user-specific dialogue context	Indirect personalization through enriched language understanding
Challenges	Volatility, context loss after session ends	Memory management, storage, privacy, and retrieval latency	Managing time-sensitive, user-specific data	Maintaining accuracy and relevance of stored world knowledge

conversational system could be accurate and personalized concurrently.

III. CHALLENGES IN LONG-TERM DIALOGUE COHERENCE

Ensuring long-term coherence in the conversation is one of the core issues in the design and development of generation-based chatbots. Since messages may take numerous turns or require numerous visits, the technology to trace and maintain the context of the dialogue, intentions, and preferences of the user becomes more complicated[17]. A chatbot is limited in its ability to maintain conversations over prolonged periods of time due to numerous technical and architecture-related issues. The following section describes the major issues that affect long-term coherence of the dialogue systems.

A. Contextual Drift and Forgetfulness

Contextual drift Contextual drift is the situation of gradually losing or drifting off conversational context during a dialogue. Generation models based on generation, especially using context windows of a fixed length, might lose important details of previous dialogue turns leading to inconsistent or off topic responses. As there is no secure system to store and revisit previous contacts, chatbots may forget the discussions, the preferences of the users, or previous arrangements[18]. This is further worsened during prolonged dialogue, where coherence of the topic, and user interest, is maintained by the proper modeling of the context. Inability to remember does not only diminish the perceived level of intelligent system but also interferes with the accomplishment of tasks and satisfaction of users.

B. Personalization Over Multiple Sessions

The key to good personalization is to have the chatbot keep and use user data in subsequent sessions. Nonetheless, a large number of systems in use are currently session-isolated, they have no persistent memory nor user profiles. This restricts the possibility of the ability to customize responses depending on past histories or changing user preferences. Individualized conversation comprises of recording the history of conversation, modeling user behavior as well as revising the profile on the fly, and, at the same time, respecting privacy and data integrity. The absence of cross-session personalization leads to repeated questions, generic replies, and a lack of perceived empathy, which negatively impacts user experience and trust in the system.

C. Catastrophic Forgetting in Neural Networks

This phenomenon of catastrophic forgetting in neural networks has been well-documented especially in continual learning settings. In case a chatbot is updated with new information or applied in a new task, it can forget what is already learned. Such phenomenon constitutes a great threat to a long-term coherence because it may wipe out effective patterns of dialogue, historical knowledge of the users, or expertise on the task. Chatbots on the basis of generation need to combine new information without forgetting the old information, so it needs more sophisticated training methods like elastic weight consolidation, rehearsal strategies, or modular networks. The inability to address the problem of catastrophic forgetting compromises the stability and build-up of long-term conversational skill.

D. Scalability and Storage Constraints

Since chatbots collect more and more dialogue data to personalize and trace context, there are two essential issues: scalability and storage efficiency. Comprehensive dialogue histories, user profiles and contextual embeddings require a significant amount of computational and memory resources to store. Moreover, real-time retrieval and reasoning over large memory stores can introduce latency, affecting responsiveness. Efficient indexing, compression, and memory management techniques are necessary to maintain performance while scaling to support large user bases and multi-turn interactions[19][20]. Balancing rich memory retention with system efficiency and scalability remains a complex engineering challenge in real-world chatbot deployments.

IV. PERSONALIZATION TECHNIQUES IN GENERATION CHATBOTS

Personalization is an important aspect when considering generation-based chatbots and is vital if wish to realize human-like, adaptive, and user-centered dialogues. Chatbots can personalize their interactions with the integrations of personalization techniques to make dialogues effective in terms of user end goals, satisfaction and engagement[21]. This section examine some of the most crucial aspects of personalization, how they incorporate techniques such as user profiling, context modeling, continual, and privacy-preserving learning, and their technical foundations and implementation challenges. Figure 1 discuss all the techniques listed below:

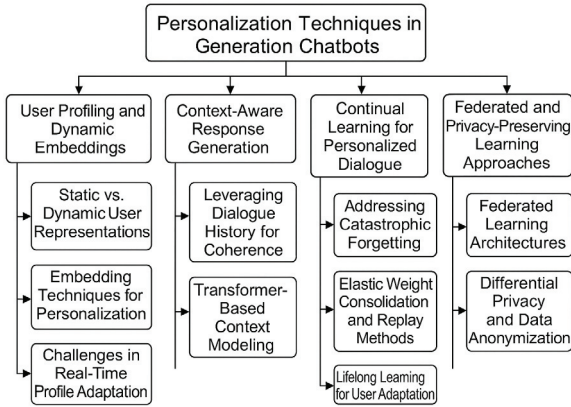


Fig. 1. Personalization techniques in generation chatbots.

A. User Profiling and Dynamic Embeddings

Personalization begins with building an accurate and adaptive profile of the user. These profiles can capture long-term preferences, behavioral patterns, demographic data, and interaction history, which enable the chatbot to adjust its responses in a user-specific manner.

1) Static vs. Dynamic User Representations

Static user profiles are generally initialized based on attributes or previous engagements, but they are fixed and never change. Dynamic user representations are constantly updated on a continuous basis at each interaction reflecting the user's ongoing needs and intentions[22]. Dynamic embeddings, specifically, give the chatbot the ability to extend their knowledge of the user by learning evolving latent representations over conversations.

B. Embedding Techniques for Personalization

The idea behind embedding-based personalization is to map user features and dialogue history to dense vector spaces, which serve as conditioning signals in the response generation process[23]. Some approaches utilize user-level embeddings, context-aware attention, and memory-based updates to vectors on the basis of user engagement, integrated into transformer architectures or sequence-to-sequence frameworks, thus conveying user-level attributes in the generated responses.

1) Challenges in Real-Time Profile Adaptation

Real-time personalization poses computational and architectural challenges. Updating embeddings or profiles on-the-fly requires efficient algorithms and low-latency inference. Moreover, maintaining consistency while avoiding overfitting to transient behaviors is critical. Balancing short-term preferences with long-term user models remains a non-trivial research problem.

2) Context-Aware Response Generation

Context-aware generation enables chatbots to produce responses that are sensitive to both the immediate dialogue context and the broader conversation history, which is critical for coherence and personalization.

3) Leveraging Dialogue History for Coherence

The dialogue history contains relevant information about past topics, user intentions, and emotional indicators. Using dialogue history prevents context drift and enables the chatbot to maintain a consistent tone and style[24]. Methods like concatenating previous utterances as a context signal and using hierarchical attention are typical ways to make sense of valuable signals from historical dialogue turns.

4) Transformer-Based Context Modeling

Both GPT, BERT are strong for modeling context because of the self-attention mechanism attached to the transformer architecture. The model's attention mechanism enables it to pay attention to long-range dependencies across dialogue turns, producing replies that are coherent and contextually grounded over longer dialogue sequences. Their pretraining on massive corpora of conversational data improves the model's ability to generalize across topics and user intents.

5) Short-Term vs. Long-Term Context Integration

Short-term context refers to the present session or maybe a few turns in the dialogue, while long term context refers to all memory held persistent across sessions. Both types of memory are important for personalization, short-term may ensure recognized relevance in the current interaction, while long-term allows for user-specific recognition and historical consistency for recall[25]. Increasingly, hybrid architectures that leverage both types of memories are being used for context-aware personalization.

C. Continual Learning for Personalized Dialogue

Continual learning allows chatbots to learn from continuous interactions without changing their previously learned knowledge, therefore enabling them to evolve alongside the user.

1) Addressing Catastrophic Forgetting

Conventional neural networks suffer from catastrophic forgetting, the ability to learn information that causes previous knowledge to be lost. This is particularly problematic in personalized chatbots which must be able to retain user-

specific preferences as they are updated over time[26]. Overcoming this problem is essential for building lifelong learning systems.

2) Elastic Weight Consolidation and Replay Methods

Elastic Weight Consolidation (EWC) prevents forgetting by preserving weights that were significant to previous tasks, and replay techniques use previously experienced dialogue samples to retrain the model from time to time. Both of these techniques allow a model to learn new information while keeping performance on old knowledge.

3) Lifelong Learning for User Adaptation

Lifelong learning frameworks modify chatbot behavior by assimilating users' feedback over long spans of time. This allows chatbots to produce a more complex[27], nuanced user profile, while improving the behavior of its response generators. The stability-plasticity trade-off -- a model being stable and retaining core behavior while being flexible -- is an ongoing challenge for personalized dialog over a lifetime.

D. Federated and Privacy-Preserving Learning Approaches

As personalization increases customer engagement, organizations must also protect user data as well. Federated learning and privacy-preserving mechanisms make sure personalization happens without sacrificing security or regulatory compliance.

1) Federated Learning Architectures

Federated learning allows for model training to happen in a decentralized manner directly on user devices, eliminating the need to send raw data to a central server. Each user device perform local updates and the server simply aggregate the model parameter updates. This approach limits risk of data breach and also enables user-specific personalization at scale.

2) Differential Privacy and Data Anonymization

In addition, differential privacy methods combat privacy concerns by injecting noise into the training process since it would not be able to memorize any specific user's data. Other data anonymization methods include hashing and tokenization methods that strip out identifiable information before it is inputs, and safeguard user privacy.

3) Trade-offs Between Privacy and Personalization

Although privacy-preserving techniques are necessary, they often come with trade-offs to personalization quality and model quality. For instance, if anonymize or add noise into data significantly, might degrade the relevance of the responses[28]. Finding an equilibrium between the accuracy of personalization and privacy protection continues to be a problem for researchers, especially in sensitive fields like healthcare and finance.

4) Centralized vs decentralised federated learning

The architectural differences between centralized and decentralized Federated Learning (FL) setups. In the centralized model, a central server coordinates and aggregates model updates, whereas the decentralized model leverages a blockchain infrastructure to orchestrate the process without relying on a central authority[24]. This structural divergence has significant implications for privacy, trust, and scalability, which compares the two approaches across critical aspects such as privacy guarantee location, noise mechanism, trust assumptions, accuracy trade-offs, and scalability.

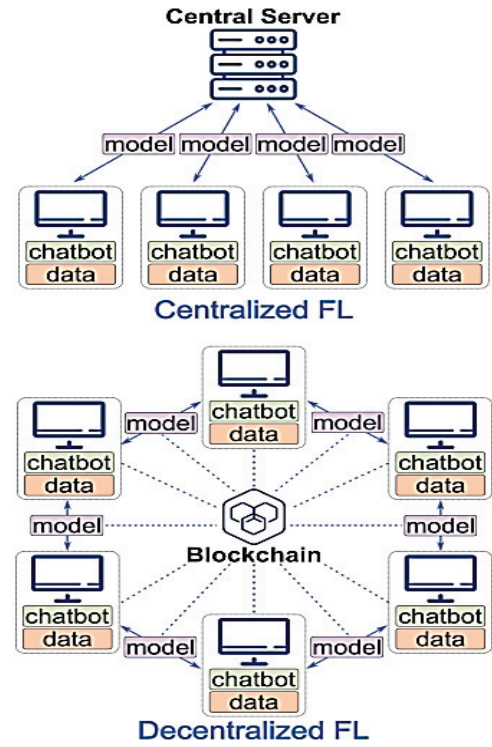


Fig. 2. Architectural differences between centralized and decentralized Federated Learning (FL).

V. LITERATURE OF REVIEW

This section presents a literature review on memory management strategies for maintaining long-term dialogue coherence and personalization in generation chatbots. It highlights developments in memory architectures (static, dynamic, episodic, personal semantic), the use of transformer-based and hybrid models, and the integration of retrieval-augmented generation (RAG). In Table 2, a succinct summary of the examined papers is presented.

Huang et al., (2025) explored the challenges in adopting text-based chatbots in e-commerce, particularly from a user experience (UX) perspective, using the Stimulus-Organism-Response (SOR) framework. A quantitative methodology was employed, surveying Indonesian e-commerce users who interact with chatbots. The study tested six hypotheses, revealing that empathy and friendliness positively influence trust, while credibility, anthropomorphism, and privacy concerns do not significantly impact trust. Trust, in turn, was found to be a critical determinant of users continued use of chatbots. These findings underscore the importance of enhancing chatbot features to better meet user needs, suggesting that improving empathy, friendliness, and contextual adaptation could lead to more trustworthy and engaging user experiences[29].

Benaddi et al. (2024) proposed a new classification of chatbots, accompanied by an exhaustive study of their evolution. employed a specific search query to find pertinent publications in the Web of Science and Scopus databases. The results of the study are summarized in an exhaustive table, listing the different contexts in which chatbots are used, as well as the underlying technologies and techniques that power their functionalities. This report offers a thorough summary of developments and patterns in the field of chatbots for tourism,

highlighting current failings and opportunities for future developments[30].

Wang et al. (2024) suggested TP-GPT, an intelligent online chatbot that uses a vast real-time traffic database to monitor and manage transportation in an effective, personalised manner. Using transportation-specific prompts, Chain-of-Thought prompting, few-shot learning, multi-agent collaboration strategy, and chat memory, the novel framework uses the contextual and generative intelligence of language models to produce precise SQL queries and natural language interpretations. A rigorous traffic-analysis benchmark shows that method performs better than state-of-the-art baselines like GPT-4 and PaLM 2, Trans-Query[31].

Ahmed, Khan and Munir, (2024) the chatbot's design by emphasizing context management in dialog-based talks and the few-shot learning problem. Initially, suggest a new hybrid intent and slots transformers (HIST) paradigm to address the few-shot learning issue with chatbots. In order to classify intent and extract slots, the HIST chatbot architecture integrates conditional random field algorithms with transformers, self-attention mechanisms, and a bigated recurrent unit. Second, provide a hybrid interaction approach for slots mapping and efficient conversational context management in order to handle dialogue management[32].

Zhang and Huang, (2024) provided insightful information about the useful advantages of LLM-based language learning resources, with a focus on vocabulary growth. LLM-based chatbots prove to be useful tools for language learning. It highlights how crucial it is for teachers to comprehend how these technologies may be used to teach L2 vocabulary and promotes the use of innovative teaching strategies that make use of these resources[33].

Shukla et al., (2023) formulated a comprehensive framework that harnesses the potential of these models and expertise of multiple human experts to enhance the coverage and correctness of information in a domain-restricted knowledgebase. This framework enables multiple domain-specific experts within a larger organization to improve a knowledgebase by querying it using a chatbot. The open-source LLMs that we used to test the proposed framework were limited to 32 GB GPU memory. A precision reduction strategy determined by QLoRA was used to fine-tune them. To our knowledge, this will be the first study to employ LLMs to cover knowledgebases[34].

Table 2 will provide an overview of the literature research with an emphasis on the subject of memory management, dialogue coherence, personalizing strategies, the main findings, and limitations as well as future directions in generative chatbots.

VI. CONCLUSION AND FUTURE WORK

Successful memory management is one of the crucial points on the way to long-term consistency and individualization of generation-based conversational agents. Since dialogue systems are still developing and gaining prominence in human-computer interaction, the capacity to maintain contextual continuation and grow with the preferences of the user, is important in the provision of a consistent and appealing user experience. This paper has given an elaborate description of different types of memory architecture, the significance of short-term memory system and long-term memory system, and critical problems of contextual drift, catastrophic forgetting and issues surrounding scalability. Also, the study has focused on personalization techniques, such as user profiling, lifelong learning, and privacy-preserving protocols, such as federated learning.

TABLE II. COMPARATIVE ANALYSIS OF LITERATURE REVIEW BASED ON MEMORY MANAGEMENT IN GENERATIVE CHATBOTS

Reference	Study Focus	Strategies	Key Findings	Challenges	Future Directions
Huang et al., (2025)	User experience (UX) in e-commerce chatbots	Stimulus-Organism-Response (SOR) framework; quantitative survey	Empathy and friendliness boost trust; trust predicts continued use	Credibility and anthropomorphism don't significantly impact trust	Improve empathy, friendliness, and contextual adaptation in chatbots
Benaddi et al., (2024)	Classification and evolution of tourism chatbots	Literature review using Web of Science and Scopus; contextual analysis	Provides a taxonomy and functional analysis of chatbot use in tourism	Fragmentation of chatbot applications; lack of unified framework	Address gaps and inconsistencies in chatbot design and implementation in tourism
Wang et al., (2024)	Intelligent transportation surveillance using chatbots	Chain-of-Thought prompting, multi-agent cooperation, chat memory, and the TP-GPT framework	Performs better on traffic-related tasks utilising domain-specific techniques than GPT-4 and PaLM 2.	Real-time adaptation to dynamic traffic conditions	Broaden application of TP-GPT to other real-time infrastructure and surveillance systems
Ahmed, Khan & Munir (2024)	Chatbot architecture for few-shot learning and dialogue management	HIST model (Hybrid Intent and Slots Transformers); hybrid slot mapping strategy	Enhances intent classification and slot extraction; improves context management	Managing conversational flow with minimal examples	Expand HIST to multilingual and multi-intent chatbot scenarios
Zhang & Huang (2024)	Chatbots for vocabulary development in language learning	Integration of LLMs in L2 teaching; strategic teaching methods	Chatbots help enhance vocabulary retention in second language learners	Teacher awareness and effective integration of LLMs	Encourage adoption of LLMs in broader L2 educational contexts
Shukla et al., (2023)	Domain-specific knowledgebase enhancement with LLMs	Human-expert collaboration; open-source LLMs; QLoRA-based fine-tuning	Improves accuracy and coverage of domain knowledgebases; supports low-resource LLMs	Hardware constraints (GPU memory); alignment with domain-specific goals	Expand to multi-domain and real-time knowledge integration with enhanced LLMs

The results help reinforce the point that effective memory mechanisms are not only essential to maintain coherency between dialogues, but they are also important to enable purposeful personalization by storing user-contained information. Going forward, a future study ought to delve into the aspect of hybridizing the memory structures which integrates both the long and short memory operations and introduce the emotional intelligence to make it more human like. The creation of safe and scalable individualization methods should also be highlighted. Real-time, multi-session application and testing will be needed in order to ensure the efficiency and flexibility of these techniques in dynamic conditions.

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